

* Home Assignment - 1 *

* Fundamentals of Research (What is Research?)

1] What are the following Terms:-

Discovery, Research, Invention, Innovation, Novelty, Creativity hence, distinguish between them with suitable example.

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- Discovery:-

Finding something that already exists but was previously unknown.

Eg:- Discovering a new planet or new species.

- Research:-

Systematic investigation to gain knowledge or understanding.

Eg:- Conducting a study on the effects of climate change.

- Invention:-

Introducing new ideas or methods.

Eg:- Developing a new teaching method or a new business model.

- Novelty:-

Being new, original, or different.

Eg:- A new style of clothing or a new genre of music.

- Creativity:-

The ability to generate new ideas or solutions.

Eg:- Writing a poem or painting a picture.

2] Justify the impact of Research on society as well as on Socioeconomic development with example.



Research has a profound impact on society & socioeconomic development. Here are some examples:-

- Scientific Research:-

- Medical research leads to development of new treatments & cures to diseases.
- Agriculture research improves crop yields & food security.
- Environmental research helps in understanding & addressing climate change.

- Technological Research:-

- Technological advancements lead to new products & services, creating jobs & economic growth.
- Innovations in communication technology improve connectivity & global interaction.
- Development of sustainable technologies helps in preserving the environment.

3] Justify the important role of Research components such as Tools, Techniques like Mathematical Modelling, Algorithms & domains with example.

→ Research components are essential for conducting effective research. Here are some examples:-

- Tools:-

- Statistical tools help in analyzing data & drawing conclusions.
- Laboratory equipment is used for conducting experiments & collecting data.

- Techniques:-

- Mathematical modeling helps in understanding complex systems & predicting outcomes.
- Algorithms are used for solving problems & making decisions.

- Domains:

- Different research domains (eg:- Biology, Physics, Engineering) have their own specific tools, techniques & methodologies.

- Example:-

A researcher studying the spread of disease might use mathematical modeling to predict its trajectory, statistical tools to analyze data on infection rates, & laboratory equipment to test for the virus. The interdisciplinary approach highlights importance of various research components in addressing complex problems.